Exercise 01: Introduction

Task 1: Schema architecture

You are tasked with developing a management system for the examination grades at the TH Rosenheim. There are only two types of users of this system: students, who are of course only allowed to see their own grades, and professors, who enter the grades for the lectures they give.

a) What could the two *external* schemas look like for these two user groups? Draw possible tables as a solution. Remember: an external schema is the output, that a database generates for a specific use case or app. **(5 Points)**

Schema for a student:

Lecture	Semester	Grade
Programming 3	WS23	1.7
Databases	WS23	1.3

Schema for a Professor:

Lecture	Semester	Matriculation Number	Grade
Programming 3	WS23	12345678	1.7
Databases	WS23	23456789	1.3

b) What could a *conceptual* schema look like? Draw possible tables as a solution. Remember, a conceptual schema is how we, as the database developers, structure the data in tables or relations. **(5 Points)**

Alternative 1: A flat-file-database

Data:

Lecturer	Lecture	Semester	Matrikulation Number	Grade
Höfig	Databases	WS23	12345678	1.2
Hüttl	Programming 3	WS24	23456789	1.3
Breunig	Databases	WS25	34567890	1.7

Alternative 2: two tables

Lectures:

Lecturer	Lecture	Semester
Höfig	Databases	WS23
Hüttl	Programming 3	WS24
Breunig	Databases	WS25

Grades:

Matrikulation Number	Lecture	Semester	Grade
12345678	Databases	WS23	1.2
23456789	Programming 3	WS24	1.3
34567890	Databases	WS25	1.7

c) What could an *internal* schema look like? Answer the question in one sentence. Remember: an internal schema is how the database internally stores the data.
You could choose a tree structure in which to store the data.

Task 2: Set up your environment

The prompt could be “create a query for the given file in tsql that provides the average grades for all students. the possible grades are 1.0, 1.3, 1.7, 2.0, 2.3, 2.7, 3.0, 3.3, 3.7, 4.0 and 5.0. round the average grades accordingly”

Task 3: Problems with data redundancy

You live in a shared apartment with two other students, Julia and Christian. Each person has their own room, and in addition you have a shared living room and kitchen. Because you get on well with each other, you do a lot of things together, cook together, celebrate together and also cook for each other. So there's a lot going on in your shared apartment!

a. List three very specific problems that this scheduling process can cause for the three of you.

   Julia wants to celebrate her birthday on 1 December, and she has told you and Christian. You and Julia have entered it in your smartphones, but Christian forgot to enter it in his calendar and invites 10 friends to watch football on 1 December. The apartment gets quite full, and Julia is cross. (Problem: forgetting changes).

   Julia is tired of her Android smartphone and buys an iPhone. Unfortunately, the iPhone calendar cannot access the data in the Android calendar. Christian helps her to transfer the data, but the two of them take 5 hours to do so and in the end they are both fed up. (Problem: data is more valuable and durable than applications).

   Michael, one of Christian’s friends who is studying political science, comes to visit and wants to look up something quickly on the living room PC. While doing so, he gets a virus that reformats the PC. All of Christian’s calendar data is lost. (Problem: data protection and data security are hard to ensure)

b. Which solution would you suggest?

   Use of a shared calendar (=database), which all three can access.

Task 4: Data independence

Your management system from Task 3 is very successful, so the Examinations Office contacts you and would also like to be able to access your database as a third user group. The Examinations Office needs access to all data, so you create a third suitable external schema for this purpose.

a) Explain what *logical data independence* means here. **(2 Points)**
It means that you do not have to change the two external schemas for the access by students and professors, even if a new external schema (for the Examinations Office) must definitely be added here, which will probably also lead to changes in the conceptual schema (e.g. it may be that the Examinations Office also requires the rooms and the invigilators of each examination, and these must naturally then be stored somewhere in the conceptual schema).

b) Due to the numerous large accesses by the Examinations Office, your database becomes very slow. So you decide to optimise the internal representation. Explain what *physical data independence* means here. What significant advantage does this have? **(2 Points)**
It means that you optimise the database by making changes to the internal schema without changing the conceptual schema (and thus the external schemas). The advantage is that you do not have to change the applications for students, professors and the Examinations Office.